Amanda
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Segmenting Singapore Neighbourhoods for Targeted Senior Digital Inclusion

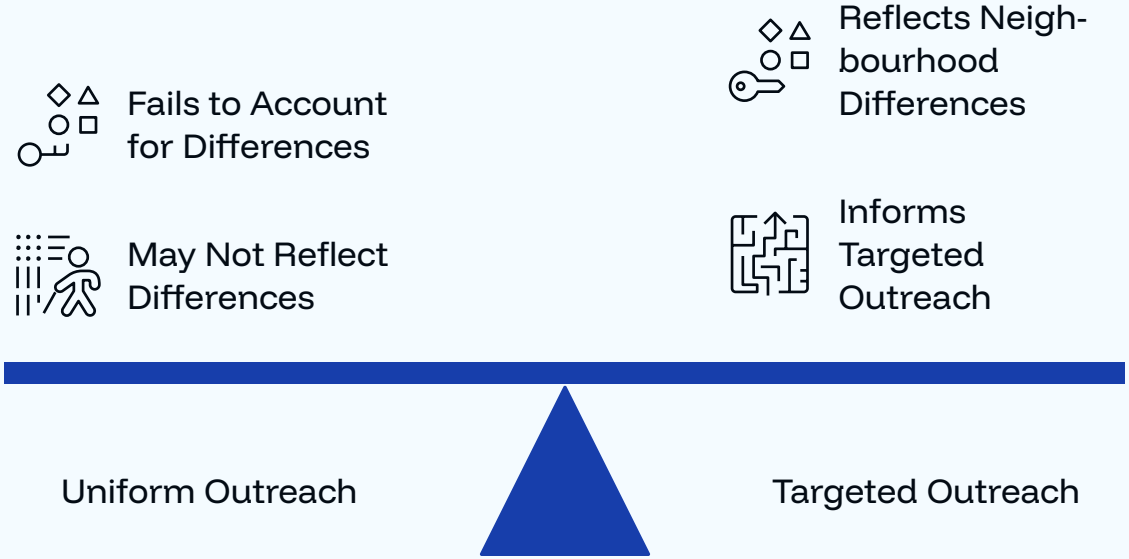
A data-driven approach to neighbourhood-specific outreach



Problem statement

What distinct neighbourhood profiles exist, and how can they inform more targeted senior digital-inclusion outreach?

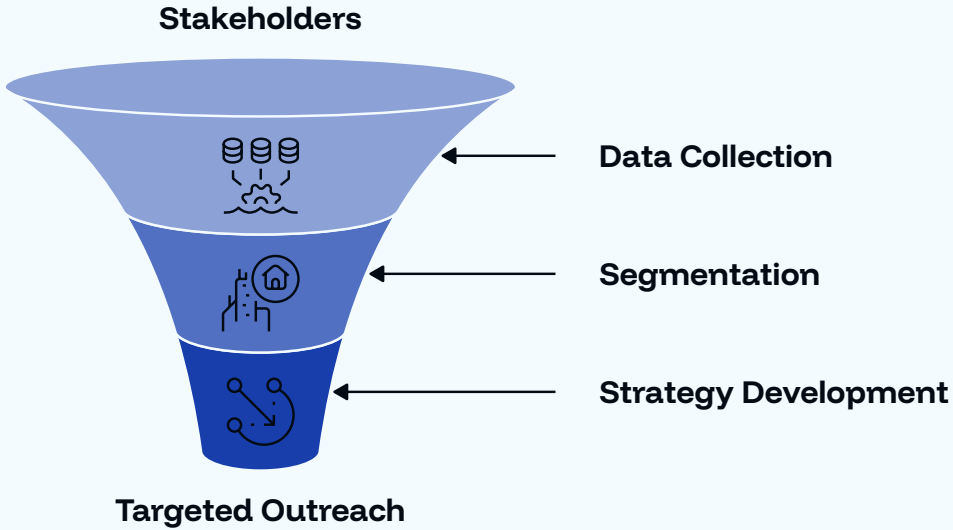
Neighbourhoods differ in senior population, Wireless@SG access, Community Club access, and Library access.



Stakeholders & Value Proposition

Use neighbourhood segmentation to support more targeted decisions on where and how senior digital-inclusion outreach could be delivered

Stakeholders include government agencies and community organisations supporting seniors who may require digital support. Limited outreach resources need to be deployed effectively.



Building a neighbourhood-level view

Analysis integrates 5 primary datasets to build a comprehensive view of 232 subzones

Public Libraries

Essential infrastructure for digital access and learning

Wireless@SG Hotspots

Public WiFi availability and proximity

Census 2020 Population

Granular demographic data on seniors aged 65+



Community Clubs

Key physical touchpoints for community engagement

URA Subzone Boundaries

Geographic framework for neighbourhood analysis

Source: data.gov.sg — SingStat, URA, IMDA, People's Association, National Library Board

Data cleaning & Feature engineering

8 features x 232 subzones



Infrastructure Proximity

- Distance to nearest Wireless@SG hotspot
- Distance to nearest Community Club
- Distance to nearest library

Infrastructure Availability

- Wireless@SG hotspot count
- Community Club count
- Library count

Demographic

- Senior population aged 65+
- Senior percentage

EDA 1: Neighbourhoods differ in both senior population & infrastructure accessibility

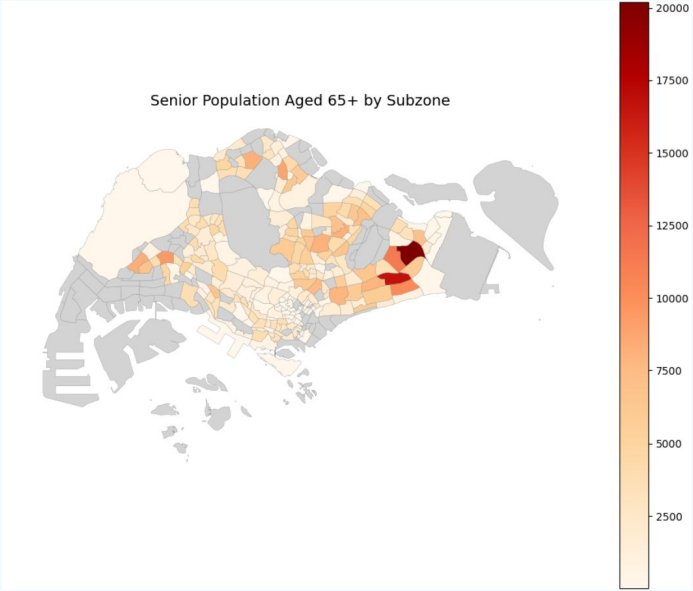
Senior Population

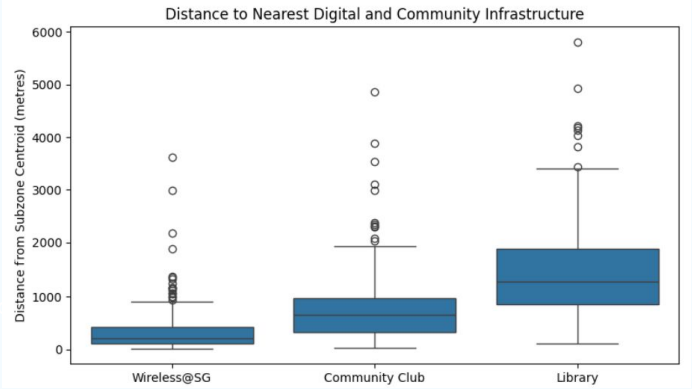
- Senior populations vary substantially across subzones
- Most subzones have smaller senior populations, while a few have much larger senior populations

Infrastructure Accessibility

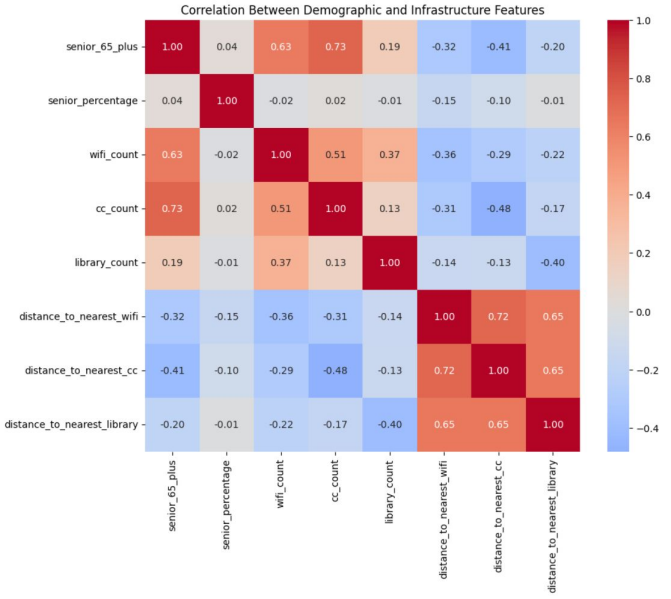
- Infrastructure availability and proximity also vary across subzones
- Wireless@SG is generally more accessible, while libraries are fewer and typically further away

Neighbourhood needs cannot be understood from senior population alone, infrastructure context matters too.





EDA 2: Subzones with larger senior populations tend to have more WiFi hotspots and Community Clubs



+0.63

Senior population ↔ WiFi

High senior populations are strongly correlated with better infrastructure access.

+0.73

Senior population ↔ Community

High senior population does not equal poor infrastructure access.

Modelling: Scale, PCA, 4 Clusters

Why Standard Scaler?

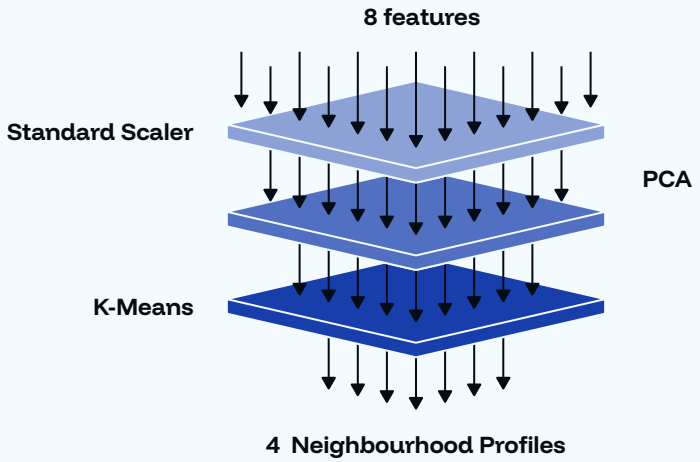
- Both PCA and K-Means are sensitive to scale
- Without Standardisation, variables with larger numerical values would have greater influence simply due to large numbers

Why Principal Component Analysis (PCA)?

- Reduce redundancy among correlated infrastructure features
- 8 to 4 principal components
- ~85% variance retained

Why 4 Clusters?

- Elbow method + silhouette analysis + interpretability
- k = 4 | Silhouette score: 0.358



Note: k=6 achieved the highest silhouette (~0.368); k=4 (~0.358) was selected for comparable separation, elbow support and greater interpretability.

Result: Four Neighbourhood Profiles Emerged

- Similar senior populations can have different infrastructure profiles
- Clusters 0 and 3 have similarly high senior populations (~4,513 vs ~4,397) but differ in the type and level of infrastructure provision

Cluster 4 (24)

- **High seniors**
- Strong WiFi & library provision
- Reasonable CC access

Cluster 1 (83)

- **High seniors**
- Good WiFi & CC access
- Limited library provision

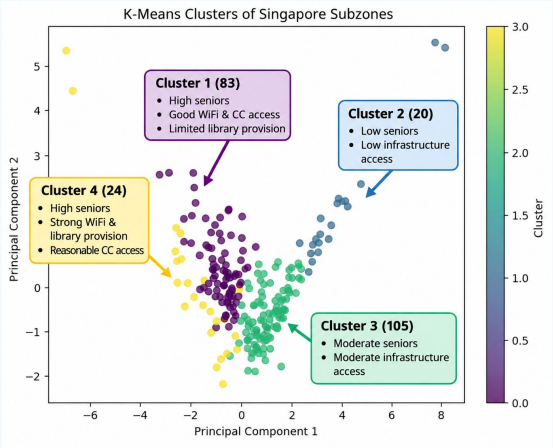


Cluster 3 (105)

- Moderate seniors
- Moderate infrastructure access

Cluster 2 (20)

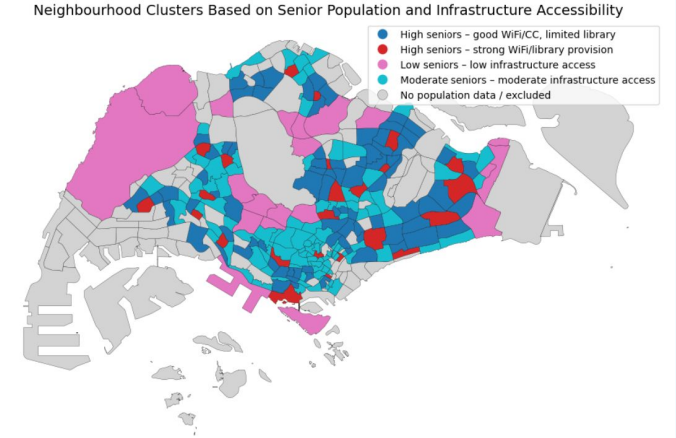
- Low seniors
- Low infrastructure access



Recommendation: From Neighbourhood profiles to Differentiated Outreach

Profiles are distributed across Singapore, highlighting the importance of neighbourhood-level targeting

Cluster	Strategy	Recommendation
1	Leverage	Leverage existing CC and WiFi touchpoints for senior digital support
2	Investigate	Validate senior-specific needs before committing additional resources
3	Maintain & Monitor	Maintain existing coverage with periodic, targeted outreach
4	Assess Utilisation	Assess whether seniors are using existing infrastructure and support programmes



Robustness, Limitations & Next Steps

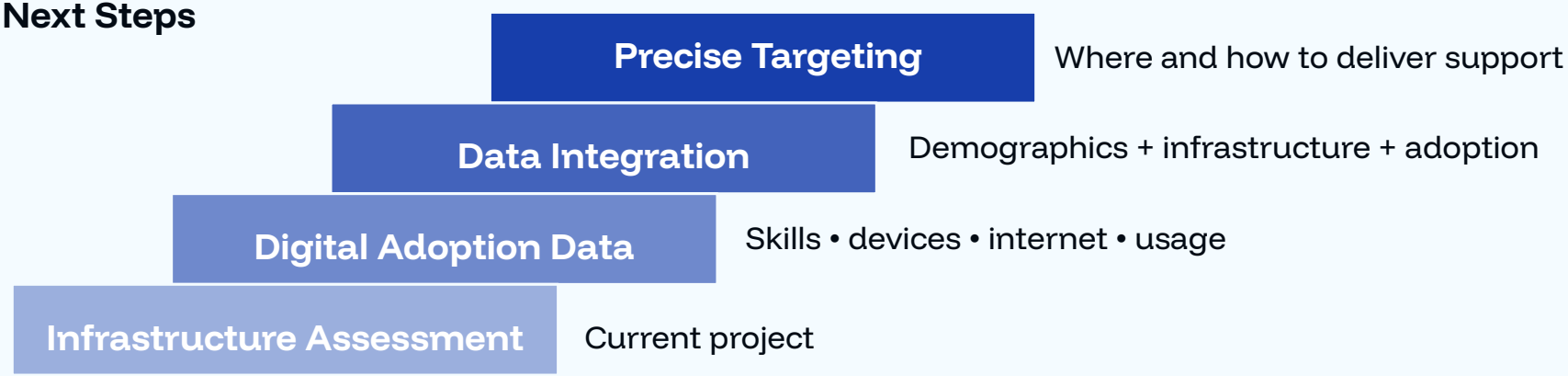
Robustness

- 98.7% cluster agreement with and without PCA
- Silhouette at k = 4, 0.358 with PCA vs 0.287 without PCA

Limitation

- Infrastructure accessibility ≠ digital inclusion

Next Steps



Thank You